

Software Engineering Department

Braude College of Engineering

Final Project – Phase A (61998)

AI Agent for Clinic Operational Management

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Abstract

Healthcare clinics operate in dynamic environments where patient arrivals, resource availability, and operational workloads continuously change throughout the day. Managing these factors efficiently is essential for reducing waiting times, improving resource utilization, and maintaining high-quality service. However, operational management often relies on manual monitoring and decision-making, which becomes increasingly challenging as clinic activity grows.

This project proposes an AI-powered operational management system for a simulated healthcare clinic environment. The system combines real-time clinic simulation with an AI Agent capable of analyzing operational conditions, detecting bottlenecks, identifying overload situations, and autonomously performing predefined management actions through a controlled tool-calling architecture.

The proposed solution focuses on operational decision support rather than medical diagnosis. By monitoring patient queues, room availability, doctor workloads, and medication inventory, the AI Agent can generate alerts, recommend actions, and provide operational summaries to support clinic staff and managers.

The project aims to demonstrate how modern AI Agent technologies can be applied to healthcare operations in a safe, explainable, and practical manner, ultimately improving clinic efficiency and resource management.

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Part 1 - Introduction

1.1 Background

In modern healthcare systems, clinics play a critical role in providing accessible medical services while ensuring efficient patient care and resource utilization. As populations continue to grow and healthcare demand increases, medical facilities face increasing pressure to manage patient flow, staff workloads, treatment rooms, and operational resources in a timely and effective manner. Within this context, efficient clinic operations have become essential for maintaining service quality, reducing waiting times, and improving overall patient satisfaction.

Despite advances in healthcare information systems, clinic operations remain highly dynamic and challenging to manage. Patient arrivals vary throughout the day, departments experience fluctuating workloads, treatment rooms continuously transition between occupied and available states, and medical resources must be monitored and allocated appropriately. These changing conditions require reception staff and clinic managers to make frequent operational decisions under time constraints, often while monitoring multiple sources of information simultaneously.

Traditionally, operational management in clinics relies heavily on human supervision and manual decision-making. Staff members are responsible for monitoring patient queues, identifying overloaded departments, allocating available resources, and responding to emerging operational issues. While this approach may be effective in smaller environments, it becomes increasingly difficult as patient volume and operational complexity grow. Human operators may experience information overload, delayed response times, and inconsistent decision-making, which can lead to longer patient waiting times, inefficient resource utilization, and reduced operational performance.

In recent years, advancements in Artificial Intelligence (AI), Large Language Models (LLMs), and AI Agent architectures have created new opportunities for intelligent decision-support systems. Unlike traditional software that primarily displays information, AI Agents are capable of analyzing complex operational environments, reasoning about current conditions, selecting appropriate actions, and interacting with predefined tools to support or automate decision-making processes. These capabilities have attracted significant interest across various industries where dynamic resource management and operational efficiency are critical.

When combined with real-time clinic monitoring systems, AI Agents provide a promising foundation for developing intelligent operational management platforms. Such systems can continuously evaluate clinic conditions, detect bottlenecks, identify overload situations, and recommend or execute predefined operational actions. By assisting healthcare personnel in managing day-to-day operations, AI-driven systems have the potential to improve resource allocation, reduce waiting times, and enhance overall clinic efficiency.

In response to these challenges, this project proposes the development of an AI-powered operational management system for a simulated healthcare clinic environment. The proposed system combines real-time clinic simulation, operational monitoring, and AI Agent reasoning to create an intelligent decision-support platform capable of analyzing clinic conditions and autonomously performing predefined operational actions. The remainder of this work presents the problem definition, reviews existing solutions, describes the proposed system architecture and development methodology, and outlines the evaluation process used to assess the effectiveness of the proposed solution.

1.2 Problem Statement

The primary problem addressed by this project is the difficulty of efficiently managing clinic operations in a dynamic environment where operational conditions continuously change throughout the day. Clinics must simultaneously manage patient flow, resource availability, and department workloads while maintaining acceptable waiting times and service quality.

Clinic personnel are responsible for monitoring several critical operational factors, including:

- Patient queues
- Waiting times
- Treatment room availability
- Doctor availability
- Department workloads
- Medication inventory levels

These factors directly affect clinic performance and often influence one another.

Current clinic management systems typically provide operational data but rely on human personnel to identify problems and determine appropriate actions. This process can be time-consuming and may result in delayed responses to overload situations, inefficient resource allocation, and reduced operational efficiency.

Therefore, there is a need for an intelligent system capable of continuously analyzing clinic conditions and assisting personnel in operational decision-making. This project investigates how an AI Agent can monitor the clinic state, detect operational bottlenecks, identify abnormal conditions, and autonomously perform predefined management actions. The objective is to improve operational efficiency, reduce manual monitoring efforts, and support more effective resource management within the clinic environment.

1.3 Existing Solutions

Healthcare organizations widely utilize Healthcare Information Systems (HIS) and Clinic Management Systems (CMS) to support daily administrative and operational activities. These systems are designed to improve information accessibility, streamline workflows, and enhance communication between healthcare professionals.

Common functionalities provided by modern healthcare management systems include:

- Electronic Health Records (EHR)
- Appointment scheduling
- Patient registration and tracking
- Resource and staff management
- Billing and administrative reporting
- Clinical documentation and record keeping

Several commercial platforms are widely adopted in healthcare environments. For example, **Epic Systems** provides a comprehensive healthcare management platform that integrates patient records, scheduling, and operational workflows. **Oracle Health (formerly Cerner Millennium)** offers solutions for patient management, clinical documentation, and healthcare administration. Similarly, **athenaOne** provides cloud-based services for patient records, scheduling, and practice management.

These systems significantly improve the organization and accessibility of healthcare information. However, their primary role is to collect, store, and present operational data to healthcare personnel. In most cases, the responsibility for monitoring clinic conditions, identifying operational bottlenecks, and determining appropriate responses remains with human operators.

Recent advances in Artificial Intelligence have introduced intelligent decision-support solutions into the healthcare domain. AI technologies are increasingly being used for applications such as medical diagnosis assistance, disease prediction, medical image analysis, and patient risk assessment. While these systems demonstrate the potential of AI in healthcare, they are generally focused on clinical decision-making rather than operational clinic management.

Consequently, there remains a gap between traditional clinic management systems, which primarily provide information, and autonomous AI-driven systems capable of actively monitoring operational conditions and assisting with day-to-day resource management. The proposed project addresses this gap by introducing an AI Agent that continuously analyzes clinic operations, identifies operational issues, and autonomously performs predefined management actions within a controlled environment. This approach extends beyond passive information management and moves toward proactive operational decision support.

Sources we can later cite in APA:

- Epic Systems documentation
- Oracle Health (Cerner) documentation
- athenahealth documentation
- Russell & Norvig, *Artificial Intelligence: A Modern Approach*
- Recent papers on AI agents and healthcare decision support (2022–2025)

1.4 Proposed Solution

This project proposes the development of an AI-powered operational management system for a simulated healthcare clinic environment.

The system will simulate a live clinic in which:

- Patients continuously arrive over time
- Departments experience changing workloads
- Treatment rooms become occupied and available
- Medication inventory changes dynamically
- Operational conditions evolve throughout the day

At the center of the system will be an AI Agent capable of:

- Monitoring clinic operations
- Detecting operational bottlenecks
- Identifying overload situations
- Selecting predefined operational tools
- Executing safe operational actions
- Generating alerts and operational summaries

Unlike traditional management systems, the proposed solution actively assists clinic personnel by analyzing operational conditions and taking predefined actions when necessary.

The project focuses solely on operational management and does not perform medical diagnosis or treatment recommendations.

1.5 Stakeholders

Reception Staff

Reception staff will benefit from automated monitoring of clinic conditions and reduced manual workload. The AI Agent can assist in identifying operational issues that require attention.

Clinic Managers

Managers will receive real-time visibility into clinic performance, operational alerts, and automated summaries that support decision-making.

Medical Staff

Doctors and clinical personnel will benefit from improved patient flow, reduced congestion, and better resource allocation.

Patients

Patients may experience shorter waiting times and more efficient service due to improved operational management.

1.6 השלמת סקר ספרות

Part 2 - Project Requirements

FR	NFR
Simulate patient arrivals at predefined time intervals.	The dashboard shall provide a clear and intuitive user interface.
Maintain patient information including department assignment, urgency level, waiting time, and treatment status.	System operations and AI actions shall be consistently recorded.
Manage clinic departments and operational states.	The AI Agent shall provide explanations for all executed actions.
Manage treatment rooms and room availability.	The system shall update operational information with minimal delay.
Track doctor availability and workload. Manage medication inventory levels.	The architecture shall support future expansion and modification.
Generate operational alerts when abnormal situations occur.	Additional departments, rooms, and users shall be supported without major redesign.
Provide a dashboard for clinic monitoring.	Access to the system shall be restricted to authorized users.
Allow users to activate the AI Agent.	
Enable the AI Agent to analyze clinic conditions.	

Part 3 - Research and Development Process

3.1 Research Phase

The research phase will focus on studying:

- AI Agent architectures
- Tool-calling frameworks
- Decision-support systems
- Healthcare operational management systems
- Simulation-based management environments

Relevant academic papers, technical documentation, and industry solutions will be reviewed in order to define the system architecture and project scope.

3.2 Development Methodology

The project will be developed incrementally using an Agile development approach.

Development will proceed through the following stages:

1. Requirements analysis
2. Database design
3. Backend implementation
4. Clinic simulation implementation
5. Dashboard development
6. AI Agent implementation
7. Integration testing
8. System evaluation

3.3 AI Agent Design

The AI Agent represents the core component of the project.

Unlike a traditional chatbot, the agent is designed as an operational AI system capable of using predefined tools to perform actions within the clinic environment.

The AI Agent will receive:

- Patient queues
- Waiting times
- Department statuses
- Room availability
- Doctor availability
- Medication inventory
- Active alerts

3.4 AI Agent Workflow

The AI Agent performs the following sequence:

Step 1 - Read Clinic State

The agent retrieves the complete operational state of the clinic.

Step 2 - Analyze Conditions

The agent identifies operational bottlenecks such as:

- Department overload
- Long waiting times
- Room shortages
- Medication shortages

Step 3 - Select Tools

The agent determines which operational tools should be activated.

Step 4 - Execute Actions

Selected tools are executed through the backend system.

Step 5 - Generate Summary

The agent generates a summary describing detected problems and executed actions.

3.5 AI Tools

- AI models from Claude/OpenAI/Gemini
- Function Calling mechanisms
- Structured JSON outputs
- Prompt engineering techniques

The AI Agent will be able to use the following operational tools:

Tool	Description
READ_CLINIC_STATE	Retrieves the complete operational state of the clinic.
OPEN_ROOM	Activates inactive treatment rooms.
CREATE_ALERT	Creates operational alerts.
SET_DEPARTMENT_STATUS	Updates department operational status.
REQUEST_MEDICATION_RESTOCK	Generates medication restocking requests.
GENERATE_SUMMARY	Produces operational summaries and reports.

3.6 Expected Outputs

The proposed system is expected to generate several outputs that support operational decision-making and clinic management:

- **Operational Alerts** - Real-time notifications for situations such as department overload, long waiting times, room shortages, and low medication inventory.
- **Department Status Updates** - Dynamic classification of departments based on their current operational condition (e.g., Normal, Warning, Overloaded, Critical).
- **AI Action Logs** - Detailed records of all actions performed by the AI Agent, including the reason for each decision and its outcome.
- **Medication Restock Requests** - Automatic requests generated when medication inventory falls below predefined thresholds.
- **Operational Summaries** - AI-generated summaries describing clinic conditions, detected issues, and actions performed.

3.7 Expected Challenges

Several challenges are anticipated during the development of the proposed system:

- **Designing Realistic Clinic Simulations** - Creating operational scenarios that accurately represent real-world clinic environments.
- **Maintaining Explainable AI Decisions** - Ensuring that all AI actions are transparent and understandable to users.
- **Preventing Unnecessary Actions** - Restricting the AI Agent to appropriate and relevant operational decisions.
- **Synchronizing Simulation and AI Analysis** - Ensuring that AI decisions are based on the most up-to-date clinic information.
- **Evaluating Agent Performance** - Defining objective metrics to measure the effectiveness of AI-generated decisions.
- **Managing Project Scope** - Balancing technical complexity with the available development time and resources.

3.8 Technologies and Tools

Component	Technology
Frontend	React
Backend	FastAPI (Python)
Database	AWS Aurora PostgreSQL
AI Integration	OpenAI/Claude/Gemini Models
API Communication	REST APIs
Version Control	Git & GitHub
Containerization	Docker